ICT 2403

VIDEO-BASED ROAD SURFACE DAMAGE DETECTION USING IMAGE PROCESSING

GROUP NO - 20

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INTRODUCTION

This is our project to process video footage of a selected road segment to detect road surface damages such as potholes, cracks, and similar surface deterioration. The detected damages are further analyzed to computationally assess their severity using image and video processing techniques.

Road surface damage occurs due to various factors, Such as continuous traffic stress, water damage, environmental factors (temperature extremes and UV radiation), poor design and construction, and natural aging of road materials.

Automated detection of road damages is important primarily because it significantly enhances public safety, improves operational efficiency and cost-effectiveness of operations, It also provides a strong foundation for modern road maintenance systems.

Image and video processing techniques play a crucial role in the automated road damage detection by automated data collection, feature extraction and damage analysis. These techniques enhance safety, support predictive maintenance, ensure standardized evaluation, and allow for faster, smarter, and more economical road maintenance management.

Further, the recorded video captures various aspects of the road surface condition, including cracks, potholes, drainage problems, uneven surface levels, and road markings, all of which have a significant impact on road users.

Thank You!